How to Use the IEEEtran LATEX Templates

IEEE Publication Technology Department

Abstract—This document describes the most common article elements and how to use the IEEEtran class with LATEX to produce files that are suitable for submission to the Institute of Electrical and Electronics Engineers (IEEE). IEEEtran can produce conference, journal and technical note (correspondence) papers with a suitable choice of class options.

Index Terms—Class, IEEEtran, LATEX, paper, style, template, typesetting.

I. Introduction

WITH the rapid development of Earth observation technologies, high-resolution remote sensing (RS) imagery has become fundamental to applications ranging from urban planning and precision agriculture to disaster assessment [1]. Remote sensing scene classification (RSSC) aims to assign categorical labels to RS images, providing a necessary foundation for subsequent structural analysis and environmental monitoring [2]. While modern deep learning models have demonstrated a sophisticated ability to extract discriminative spatial and structural patterns [3], [4], [5], [6], [7], their performance typically depends on the availability of large-scale, annotated datasets. Acquiring such large-scale annotations is often impractical due to the high costs and specialized expertise required to distinguish visually similar structures in high-resolution optical images. This annotation bottleneck poses a significant challenge for conventional data-intensive architectures, which frequently suffer from overfitting and poor generalization when trained on limited labeled samples. These limitations are particularly evident when scene classification systems need to identify novel scene categories under constrained data conditions.

Inspired by the capability of human visual cognition to learn novel concepts from minimal observations [8], few-shot learning (FSL) provides a practical paradigm to overcome the bottleneck of data scarcity [9], [10], [11], [12], [13]. The core mechanism of FSL involves extracting transferable prior knowledge from an abundantly annotated base dataset, enabling models to identify novel categories with only a limited number of labeled support samples (e.g., 1-shot or 5-shot settings). Current mainstream FSL strategies are broadly categorized into optimization-based [11], [13], [14], generation-based [15], [16], [17], and metric-based approaches [9], [10], [12]. Among these categories, metric-based approaches are widely adopted in the recent research of few-shot remote sensing scene classification (FSRSSC) due to their low computational complexity and robust performance [18], [19], [20], [21], [22]. These methods perform classification by computing the distance between query samples and class prototypes within a learned embedding space. However, existing metric-based methods excessively rely on the visual modality. Consequently, under conditions of extreme data scarcity, particularly in the 1-shot setting, the reliance on a solitary visual sample often yields unstable and non-discriminative feature representations [23].

To address these issues, researchers often incorporate textual semantics as auxiliary information [24], [25], [26], [27], [28], [29]. For instance, Xing et al. [24] present an adaptive modality mixing mechanism that dynamically fuses visual and semantic representations, effectively mitigating the unreliability of visual features under extreme data scarcity. Conceptualizing this unreliability as feature incompleteness, Zhang et al. [26] propose a prototype completion framework that uses prior semantic knowledge to restore missing visual attributes, thereby estimating more representative class centers. To bridge the semantic gap between abstract text descriptions and local visual features, Liu et al. [28] employ large language models to derive specific class entities. They progressively aggregate these fine-grained semantic patterns across different network stages to calibrate visual prototypes more accurately. Motivated by the progress of semantic-assisted FSL in general computer vision, researchers have begun investigating the use of semantic priors for the FSRSSC task [23], [30], [31], [32]. For example, Chen et al. [23] propose CNSPN, which extracts semantic priors directly from scene class names via word embeddings. These priors are then linearly fused with visual features to enhance class prototypes when training data is scarce. Additionally, Ji et al. [32] introduce learnable prompts to extract textual priors and distill them into a lightweight visual backbone, achieving robust few-shot classification without heavy inference costs.

Although semantics-assisted FSRSSC methods have shown significant progress, several critical challenges remain. High-resolution remote sensing images typically contain a dense mix of diverse land covers. Because of their extensive spatial coverage, target objects are often submerged in highly heterogeneous backgrounds alongside co-occurring irrelevant structures. As illustrated in Fig. 1, while the primary target of the first image is the stadium, it is heavily surrounded by irrelevant objects such as buildings, roads, and trees. This structural interference represents a pervasive issue across remote sensing imagery.

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Fig. 1. Complexity of background in high-resolution remote sensing scenes. The core targets are easily overwhelmed by dense urban clutters or visually camouflaged.

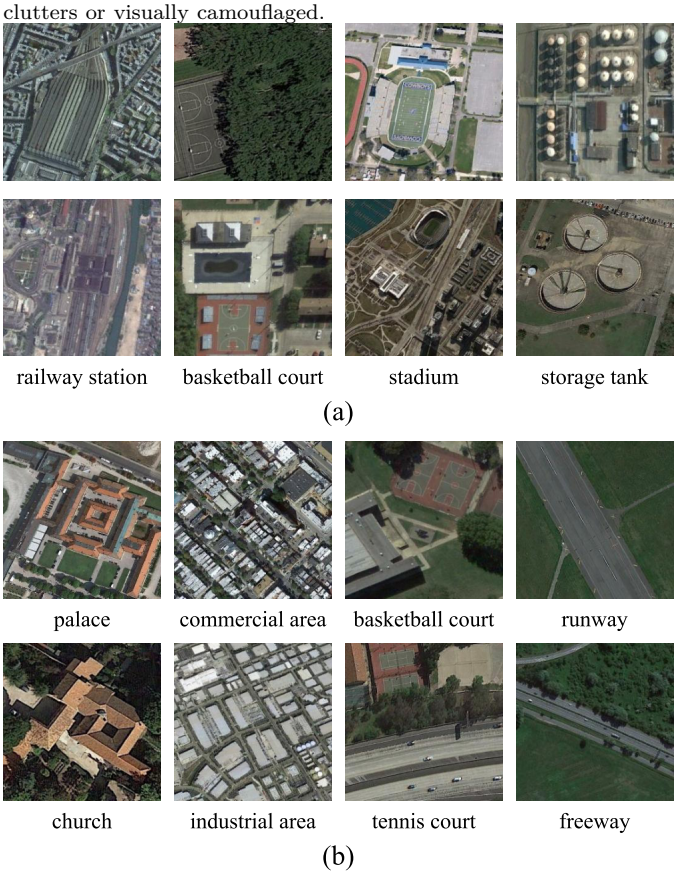

Fig. 2. This is the caption for Figure 2.

However, current semantics-assisted FSRSSC approaches [23], [30], [32] typically adopt a late-fusion strategy that lacks active semantic guidance during early feature extraction. Consequently, the weak visual signals of small targets are often diluted by complex backgrounds before they can be effectively integrated with textual features. Furthermore, remote sensing images exhibit substantial intra-class variance and inter-class similarity. As demonstrated in Fig. 2, variations in illumination, imaging viewpoints, and spatial scales cause scenes within the same semantic category to display distinctly different visual appearances. Simultaneously, scenes from different categories frequently share similar surface appearances and geometric layouts, leading to potential categorical confusion. Under extreme data scarcity, this inherent variance and similarity becomes a critical bottleneck because models lack sufficient training observations to differentiate reliable, category-specific patterns from deceptive visual overlaps. Despite these challenges, existing semantics-assisted FSRSSC methods [23], [30], [32] predominantly rely on mean-based strategies for prototype construction. This practice risks incorporating outliers, which causes prototype representation shifts and ultimately induces misclassification, particularly for visually similar scenes.

To address the above challenges, this article proposes a novel multimodal framework for few-shot remote sensing scene classification. Targeting the semantic deviation common to large language models in the remote sensing domain, we introduce the Visual Grounded Semantic Refinement (VGSR) module. This module directly anchors to visual evidence in remote sensing images to suppress semantic hallucinations during text generation, providing reliable prior knowledge for the entire network. Existing methods often perform late fusion exclusively at the classifier level, where background noise and target features remain heavily entangled. To overcome this structural limitation, we embed the Dual Cross-Modal Attention (DCMA) module deep within the intermediate stages of the visual backbone. Using the VGSR-generated text descriptions as a semantic prior, DCMA executes cross-modal interactions across spatial and channel dimensions. This actively suppresses complex background clutter during feature extraction and enhances discriminative target features, avoiding the loss of spatial layout information typically caused by global pooling operations. Traditional mean aggregation strategies are highly sensitive to outliers and prone to prototype shift. To resolve this critical weakness, we propose the Adaptive Prototype Construction Module (APCM). APCM dynamically assigns adaptive weights to each sample by jointly evaluating the intra-class compactness and inter-class repulsion of support set samples in the metric space. This mitigates interference from noisy samples and broadens decision boundaries between easily confused categories, establishing a robust metric baseline for scenarios with high inter-class similarity. For the inference stage, we introduce the Compactness-Aware Prototype Fusion (CPF) strategy. CPF quantitatively evaluates the distribution uncertainties of purely visual and semantically enhanced prototypes to achieve an adaptive dynamic fusion of both modalities, further improving the reliability of classification decisions under extreme data scarcity. The main contributions of this article can be summarized as follows:

1) A novel multimodal framework for FSRSSC is proposed. The contributions of this research center on a comprehensive multimodal framework that systematically resolves the severe challenges of background clutter, significant intra-class variance, and high inter-class similarity in remote sensing imagery, establishing a robust paradigm for classification tasks under extremely data-scarce conditions.
2) We introduce the VGSR and DCMA modules for active feature modulation. Within this framework, the developed VGSR and DCMA modules introduce visually anchored semantic priors and advance cross-

modal interactions to the feature extraction stage. They achieve explicit background noise suppression and overcome the representation bottlenecks associated with traditional late fusion methodologies.

3) We design the APCM and transductive CPF modules for robust metric learning. Concurrently, the APCM and CPF modules allocate dynamic weights based on dual intra-class and inter-class constraints in the metric space and perform uncertainty-based adaptive prototype fusion. Together, they prevent outlier-induced prototype distortion and significantly enhance classification robustness across highly similar categories.
4) Comprehensive experiments demonstrate superior performance. Comprehensive experimental evaluations on multiple mainstream few-shot remote sensing public datasets demonstrate that the proposed framework significantly outperforms existing state-of-the-art approaches, thoroughly verifying the effectiveness of the core modules in processing complex remote sensing scenes.

The rest of this article is organized as follows. Section ?? discusses the related work of FSRSSC. Section ?? presents the proposed method. Section ?? evaluates the proposed method on three mainstream remote sensing datasets. Finally, Section ?? concludes the paper.

II. Related Work

The templates are intended to approximate the final look and page length of the articles/papers. Therefore, they are NOT intended to be the final produced work that is displayed in print or on IEEEXplore®. They will help to give the authors an approximation of the number of pages that will be in the final version. The structure of the LATEXfiles, as designed, enable easy conversion to XML for the composition systems used by the IEEE's outsource vendors. The XML files are used to produce the final print/IEEEXplore® pdf and then converted to HTML for IEEEXplore®. Have you looked at your article/paper in the HTML version?

III. Method

A. Overview

To tackle the intrinsic challenges of complex background interference, substantial intra-class variance, and high inter-class similarity in high-resolution remote sensing (RS) imagery, we propose a novel and comprehensive multimodal framework for few-shot remote sensing scene classification (FSRSSC). Formally, given an N-way K-shot task episode comprising a support set $\mathcal{S}$ and a query set $\mathcal{Q}$, the proposed architecture aims to learn a highly discriminative and robust metric space. The overall pipeline of our framework is illustrated in Fig. ??. Specifically, the framework consists of four meticulously designed components. First, the Visual Grounded Semantic Refinement (VGSR) module is introduced to synthesize high-quality, visually anchored class descriptions via a Large Language Model (LLM), thereby suppressing semantic hallucinations and providing reliable prior knowledge. Subsequently, to overcome the representational bottlenecks of traditional late-fusion strategies, the Dual Cross-Modal Attention (DCMA) module is embedded deeply within the intermediate stages of the visual backbone. Guided by the VGSR-generated textual priors, DCMA executes multi-scale cross-modal interactions along both spatial and channel dimensions, which actively suppresses redundant RS background clutters and highlights discriminative foreground targets. After obtaining the enhanced visual representations, the Adaptive Prototype Construction Module (APCM) dynamically constructs representative class centers in the metric space. By jointly evaluating the intra-class compactness and inter-class repulsion, APCM adaptively allocates weights to support samples, thereby mitigating the interference from outliers and broadening the decision boundaries for easily confused categories. Finally, during the meta-test inference phase, a Compactness-Aware Prototype Fusion (CPF) strategy driven by uncertainty estimation is seamlessly integrated. CPF quantitatively evaluates the distribution uncertainties to achieve an adaptive dynamic fusion of purely visual and semantically enhanced prototypes, further rectifying the prototype shift typically caused by extreme data scarcity. The final classification decision is determined by computing the cosine similarity between the query features and the rectified prototypes, and the entire framework is optimized end-to-end using the standard cross-entropy loss.

B. Visual Grounded Semantic Refinement

A critical limitation of existing semantics-assisted methods is that large language models (LLMs) often generate generic descriptions or semantic hallucinations that lack grounding in complex remote sensing realities. To overcome this semantic ambiguity and bridge the modality gap, we introduce the Visual Grounded Semantic Refinement (VGSR) module to synthesize highly discriminative, visually anchored textual priors. The Visual Grounded Semantic Refinement (VGSR) module is designed to generate highly discriminative and visually anchored textual priors. Given a specific remote sensing scene category c and its corresponding K-shot support images $\mathcal{S}_c$, this module leverages a Vision-Language Model (VLM) to execute a three-stage structured prompting protocol to obtain refined category descriptions. The forward process proceeds as follows.

First, in the Semantic Prior Generation phase, the VLM synthesizes an intrinsic description of category c utilizing its rich remote sensing domain knowledge, focusing on invariant attributes such as typical geometric structures, spatial configurations, and textural signatures.

Second, during the Visual Evidence Extraction phase, the model analyzes the provided support set $\mathcal{S}_c$ to identify observable visual features, explicitly documenting factual cues including structural patterns and dominant textures present in the actual images.

Third, in the Semantic Refinement phase, the module performs a cross-modal intersection. It isolates features that are consistent between the semantic prior and the visual observations, while filtering out instance-specific anomalies such as transient illumination, localized background clutter, and sensor artifacts.

Through these three stages, the VGSR module outputs a concise and robust single-sentence text T_c that characterizes the generalized visual features of the category. Subsequently, this text is fed into a pre-trained text encoder to extract the text embedding $E_c \in \mathbb{R}^D$, which serves as the semantic guidance for the succeeding Dual Cross-Modal Attention (DCMA) module. The adopted prompt template is presented as follows.

Prompt for Visual Grounded Semantic Refinement

Task: Generate a concise, class-level description for {class_name} in remote sensing imagery, utilizing the category label and a set of representative images.

Step 1: Semantic Prior Generation

Synthesize a description of {class_name} grounded in remote sensing domain knowledge. Focus on invariant attributes such as typical geometric structures, spatial configurations, and textural signatures.

Step 2: Visual Evidence Extraction

Analyze the provided support images to identify observable visual features. Document factual cues, including structural patterns, spatial layouts, and dominant textures.

Step 3: Semantic Refinement

Refine the description by isolating features that are consistent between the semantic prior and visual observations. Filter out instance-specific anomalies, such as transient illumination, localized background clutter, or sensor artifacts.

Output: Provide a single, concise sentence that robustly describes {class_name} in remote sensing imagery, emphasizing generalized visual characteristics.

Class Name: {class_name}

Images: [Provided below]

The primary technical advantage of this mechanism is its ability to effectively eliminate the inherent semantic hallucinations of LLMs and fundamentally resolve the modality contradiction between visual features and textual descriptions. By establishing a reliable and visually anchored textual prior, it equips the subsequent cross-modal feature fusion with strong robustness against complex background interference, maintaining high discriminability even under extreme data scarcity.

C. Dual Cross-Modal Attention

High-resolution remote sensing imagery is inherently characterized by dense and complex backgrounds, where the weak visual signals of discriminative targets are easily overwhelmed by co-occurring irrelevant structures. While recent semantics-assisted few-shot methods [23], [32] attempt to alleviate this by introducing textual information, they predominantly rely on late-fusion paradigms, such as simple concatenation at the classifier or prototype level. This strategy is structurally sub-optimal, as the visual features of targets and background noise are already heavily entangled within the visual backbone, and crucial spatial layout information is often irreversibly discarded by global pooling operations prior to the fusion stage. To break this representational bottleneck, we propose the Dual Cross-Modal Attention (DCMA) module, which embeds textual priors deep within the intermediate visual backbone to actively suppress background clutters and purify foreground targets simultaneously across spatial and channel dimensions.

The Dual Cross-Modal Attention (DCMA) module aims to actively suppress complex background clutters and purify foreground targets in remote sensing imagery through deep visual-textual integration. Given the intermediate visual features $\mathbf{X} \in \mathbb{R}^{B\times N\times C}$ from the visual backbone (where N is the number of spatial patches and C is the channel dimension), along with the global text embedding $\mathbf{E}_c \in \mathbb{R}^{B\times 1\times C}$ generated by the VGSR module, the forward process of DCMA consists of three primary interaction and fusion steps.

First, in the Spatial Cross-Attention branch, the model formulates the visual features $\mathbf{X}$ as the query, and the text embedding $\mathbf{E}_c$ as both the key and value for a multi-head attention mechanism. This branch deliberately removes the conventional internal residual connection while retaining layer normalization, ensuring that the textual prior directly guides the model to focus on discriminative target regions along the spatial dimension.

Second, in the Channel Cross-Modal Modulation branch, the model performs global average pooling on $\mathbf{X}$ to extract a visual context vector, which is then concatenated with the text embedding $\mathbf{E}_c$. The joint representation is fed into a multi-layer perceptron (MLP) to generate channel-wise attention weights, which are subsequently used to dynamically modulate the original visual features $\mathbf{X}$ along the channel dimension via a residual addition.

Third, a Mean-guided Adaptive Interaction Module (MAIM) fuses the outputs of the dual branches. It computes the mean of the dual-branch features to generate a global weight token, which queries the branch outputs via a pure cross-attention mechanism. Adaptive fusion weights are calculated through softmax normalization, culminating in a weighted sum of the spatial and channel features.

Finally, the fused cross-modal representation is injected back into the main visual backbone through a learnable gated residual mechanism, thereby outputting the enhanced visual representation for the subsequent stages. This deep interaction mechanism overcomes the severe entanglement of target features and background noise inherent in traditional late-fusion approaches. Furthermore,

the learnable gated residual injection guarantees initial gradient stability, effectively preventing visual representations from being overwhelmed by textual priors.

D. Adaptive Prototype Construction Module

In standard metric-based few-shot learning, class prototypes are typically constructed by simply applying average pooling to the feature representations of support samples. However, high-resolution remote sensing scenes intrinsically exhibit enormous intra-class variance and high inter-class similarity, implying that a randomly sampled support set under extreme few-shot conditions often contains visually anomalous noisy instances or outliers. Consequently, the traditional mean-based aggregation strategy is highly susceptible to these outliers, inevitably leading to severe prototype shift and blurring the decision boundaries between easily confused categories. To address this critical metric bottleneck, we propose the Adaptive Prototype Construction Module (APCM), which dynamically explores Euclidean topological relations to adaptively weight support samples and calibrate prototypes using boundary instances, thereby effectively resisting outlier interference to establish robust class centers.

The Adaptive Prototype Construction Module (APCM) is designed to overcome the vulnerability of traditional mean-based prototypes to outliers by dynamically evaluating sample topological relations within the Euclidean space. Given the enhanced visual features output by the DCMA module, we first perform global average pooling to obtain the compact feature representations $\mathbf{F}_j = \{\mathbf{f}_{j,i}\}_{i=1}^{K} \subset \mathbb{R}^C$ for the j-th class support set. The forward construction process of APCM consists of three phases.

First, in the Intra-class Compactness Evaluation, we compute the pair-wise raw cosine distance matrix $\mathbf{M}_j \in \mathbb{R}^{K\times K}$ within $\mathbf{F}_j$. This distance matrix is flattened and fed into a support controller (implemented via a multi-layer perceptron) to generate an adaptive internal weight vector $\boldsymbol{\alpha}_j = \{\alpha_{j,i}\}_{i=1}^{K}$. Accordingly, the initial prototype is calculated as $\tilde{\mathbf{p}}_j = \frac{1}{K}\sum_{i=1}^{K}\alpha_{j,i}\mathbf{f}_{j,i}$.

Second, in the Inter-class Boundary Exploration, to calibrate the prototype utilizing adjacent category information, we construct a candidate set $\mathcal{C}_j$ that strictly excludes the support samples of the current class j, and project it through a linear transformation layer $\phi(\cdot)$. By calculating the cosine distances between $\tilde{\mathbf{p}}_j$ and the projected candidate samples, the m nearest samples are selected to form the boundary set $\mathcal{B}_j = \{\mathbf{b}_{j,l}\}_{l=1}^{m}$, while the remainder constitute the non-boundary set $\bar{\mathcal{B}}_j$.

Third, in the Adaptive Prototype Rectification, we compute four sets of topological distance metrics in the Euclidean space: 1) the distances between $\tilde{\mathbf{p}}_j$ and $\mathcal{B}_j$; 2) the mean distance between $\tilde{\mathbf{p}}_j$ and $\bar{\mathcal{B}}_j$; 3) the mean distances between other initial prototypes $\tilde{\mathbf{p}}_k$ $(k \neq j)$ and $\mathcal{B}_j$; and 4) the overall mean distance between other initial prototypes and $\bar{\mathcal{B}}_j$. These four distance vectors are concatenated and fed into a rerank controller, which outputs boundary calibration weights $\boldsymbol{\beta}_j = \{\beta_{j,l}\}_{l=1}^{m}$ along with a balancing scalar $\gamma_j \in (0,1)$. Ultimately, by jointly fusing the in-class support features and out-of-class boundary features, the robust rectified prototype $\mathbf{p}_j^*$ is formulated as follows:

$$\mathbf{p}_j^* = \frac{1}{K+m}\left(\sum_{i=1}^{K}\gamma_j\alpha_{j,i}\mathbf{f}_{j,i} + \sum_{l=1}^{m}(1-\gamma_j)\beta_{j,l}\mathbf{b}_{j,l}\right) \quad (1)$$

This dynamic topological weighting mechanism mitigates the severe prototype shift caused by outliers under extreme few-shot conditions. Furthermore, the boundary-aware calibration explicitly broadens the decision boundaries between easily confused categories, yielding a highly discriminative metric space.

E. Compactness-Aware Prototype Fusion

In existing semantics-assisted few-shot remote sensing scene classification (FSRSSC) methods, the final classification prototypes are typically obtained through a static linear fusion of visual features and semantic priors. However, this rigid fusion paradigm employs fixed weight coefficients across all categories and tasks, failing to account for the dynamic variations in the relative reliability of visual and semantic modalities under different meta-test episodes. Consequently, when confronted with extreme data scarcity or severe distribution shifts, fixed linear fusion is highly susceptible to modality imbalance, which inevitably triggers irreversible prototype deviation. To overcome this representational limitation, we introduce the Compactness-Aware Prototype Fusion (CPF) strategy, which dynamically allocates fusion weights based on test-time predictive uncertainty rather than static empirical hyperparameters, thereby explicitly calibrating prototypes without requiring additional training.

The Compactness-Aware Prototype Fusion (CPF) strategy aims to dynamically calibrate class prototypes during the meta-test inference phase by quantifying the uncertainty of feature distributions, thereby alleviating prototype shift under extreme few-shot conditions without requiring additional training. Given the rectified prototypes $\mathbf{p}_j^*$ generated by APCM and the query set features $\mathbf{q} \in \mathcal{Q}$, the forward computation of CPF comprises three steps.

First, in the Initial Prediction and Uncertainty Estimation, we calculate the cosine similarity between the query sample $\mathbf{q}$ and each class prototype $\mathbf{p}_j^*$, converting it into a predictive probability distribution via the Softmax function. Based on this distribution, we introduce Shannon Entropy to quantitatively evaluate the prediction uncertainty $U(\mathbf{q})$ for each query sample.

Second, in the High-Confidence Pseudo-Prototype Construction, we filter out high-confidence query samples whose uncertainties fall below a predefined threshold. Treating these samples as pseudo-labeled instances, we compute the mean of their features to construct a query-adapted pseudo-prototype $\mathbf{p}_j^q$ for the j-th class, which

fully exploits the current data manifold distribution of the query set.

Third, in the Compactness-Driven Dynamic Fusion, we calculate a compactness-aware weight $\eta_j \in (0, 1)$ based on the average uncertainty of the high-confidence samples assigned to the j-th class. A more compact feature distribution (i.e., a lower average uncertainty) yields a larger weight η_j allocated to the pseudo-prototype.

Finally, we dynamically fuse the APCM prototype $\mathbf{p}_j^*$ and the pseudo-prototype $\mathbf{p}_j^q$ to obtain the final test-time prototype $\hat{\mathbf{p}}_j$ as follows:

$$\hat{\mathbf{p}}_j = (1 - \eta_j)\mathbf{p}_j^* + \eta_j \mathbf{p}_j^q \tag{2}$$

The ultimate classification predictions are determined by computing the cosine similarity between all query samples and the final prototypes $\hat{\mathbf{p}}_j$. This uncertainty-driven fusion strategy effectively mitigates prototype deviation and guarantees robust test-time generalization without incurring any additional training overhead.

IV. Where to get the IEEEtran Templates

The IEEE Template Selector will always have the most up-to-date versions of the LaTeX and MSWord templates. Please see: https://template-selector.ieee.org/ and follow the steps to find the correct template for your intended publication. Many publications use the IEEETran LaTeX templates, however, some publications have their own special templates. Many of these are based on IEEEtran, but may have special instructions that vary slightly from those in this document.

V. Where to get LaTeX help - user groups

The following on-line groups are very helpful to beginning and experienced LaTeX users. A search through their archives can provide many answers to common questions.

http://www.latex-community.org/
https://tex.stackexchange.com/

VI. Document Class Options in IEEEtran

At the beginning of your LaTeX file you will need to establish what type of publication style you intend to use. The following list shows appropriate documentclass options for each of the types covered by IEEEtran.

Regular Journal Article
\documentclass[journal]IEEEtran

Conference Paper
\documentclass[conference]IEEEtran

Computer Society Journal Article
\documentclass[10pt,journal,compsoc]IEEEtran

Computer Society Conference Paper
\documentclass[conference,compsoc]IEEEtran

Communications Society Journal Article
\documentclass[journal,comsoc]IEEEtran

Brief, Correspondence or Technote
\documentclass[9pt,technote]IEEEtran

There are other options available for each of these when submitting for peer review or other special requirements. IEEE recommends to compose your article in the base 2-column format to make sure all your equations, tables and graphics will fit the final 2-column format. Please refer to the document "IEEEtran_HOWTO.pdf" for more information on settings for peer review submission if required by your EIC.

VII. How to Create Common Front Matter

The following sections describe general coding for these common elements. Computer Society publications and Conferences may have their own special variations and will be noted below.

A. Paper Title

The title of your paper is coded as:

```
\title{The Title of Your Paper}
```

Please try to avoid the use of math or chemical formulas in your title if possible.

B. Author Names and Affiliations

The author section should be coded as follows:

```
\author{Masahito Hayashi
\IEEEmembership{Fellow, IEEE}, Masaki Owari
\thanks{M. Hayashi is with Graduate School
of Mathematics, Nagoya University, Nagoya,
Japan}
\thanks{M. Owari is with the Faculty of
Informatics, Shizuoka University,
Hamamatsu, Shizuoka, Japan.}
}
```

Be sure to use the \IEEEmembership command to identify IEEE membership status. Please see the "IEEEtran_HOWTO.pdf" for specific information on coding authors for Conferences and Computer Society publications. Note that the closing curly brace for the author group comes at the end of the thanks group. This will prevent you from creating a blank first page.

C. Running Heads

The running heads are declared by using the \markboth command. There are two arguments to this command: the first contains the journal name information and the second contains the author names and paper title.

```
\markboth{Journal of Quantum Electronics,
Vol. 1, No. 1, January 2021}
{Author1, Author2,
\MakeLowercase{\textit{(et al.)}}:
Paper Title}
```

D. Copyright Line

For Transactions and Journals papers, this is not necessary to use at the submission stage of your paper. The IEEE production process will add the appropriate copyright line. If you are writing a conference paper, please see the "IEEEtran_HOWTO.pdf" for specific information on how to code "Publication ID Marks".

E. Abstracts

The abstract is the first element of a paper after the \maketitle macro is invoked. The coding is simply:

```
\begin{abstract}
Text of your abstract.
\end{abstract}
```

Please try to avoid mathematical and chemical formulas in the abstract.

F. Index Terms

The index terms are used to help other researchers discover your paper. Each society may have it's own keyword set. Contact the EIC of your intended publication for this list.

```
\begin{IEEEkeywords}
Broad band networks, quality of service
\end{IEEEkeywords}
```

VIII. How to Create Common Body Elements

The following sections describe common body text elements and how to code them.

A. Initial Drop Cap Letter

The first text paragraph uses a "drop cap" followed by the first word in ALL CAPS. This is accomplished by using the \IEEEPARstart command as follows:

```
\IEEEPARstart{T}{his} is the first paragraph
of your paper. . .
```

B. Sections and Subsections

Section headings use standard LATEX commands: \section, \subsection and \subsubsection. Numbering is handled automatically for you and varies according to type of publication. It is common to not indent the first paragraph following a section head by using \noindent as follows:

```
\section{Section Head}
\noindent The text of your paragraph . . .
```

C. Citations to the Bibliography

The coding for the citations are made with the LATEX \cite command. This will produce individual bracketed reference numbers in the IEEE style. At the top of your LATEX file you should include:

```
\usepackage{cite}
```

For a single citation code as follows:

Fig. 3. This is the caption for one fig.

```
see \cite{ams}
```

This will display as: see [33]

For multiple citations code as follows:

```
\cite{ams,oxford,lacomp}
```

This will display as [33], [34], [35]

D. Figures

Figures are coded with the standard LATEX commands as follows:

```
\begin{figure}[!t]
\centering
\includegraphics[width=2.5in]{fig1}
\caption{This is the caption for one fig.}
\label{fig1}
\end{figure}
```

The [!t] argument enables floats to the top of the page to follow IEEE style. Make sure you include:

```
\usepackage{graphicx}
```

at the top of your LATEXfile with the other package declarations.

To cross-reference your figures in the text use the following code example:

```
See figure \ref{fig1} ...
```

This will produce:
See figure 3 . . .

E. Tables

Tables should be coded with the standard LATEX coding. The following example shows a simple table.

```
\begin{table}
\begin{center}
\caption{Filter design equations  ...}
\label{tab1}
\begin{tabular}{| c | c | c |}
```

TABLE I
A Simple Table Example.

Order of filter	Arbitrary coefficients e_m	coefficients b_{ij}
1	$b_{ij} = \hat{e}.\hat{\beta_{ij}}$,	$b_{00} = 0$
2	$\beta_{22} = (\ 1, -1, -1,\ \ 1,\ \ 1,\ \ 1)$	
3	$b_{ij} = \hat{e}.\hat{\beta_{ij}}$,	$b_{00} = 0$,

```
\hline
Order & Arbitrary coefficients &
coefficients\\
of filter & $e_m$ &    $b_{ij}$ \\
\hline
1& $b_{ij}=\hat{e}.\hat{\beta_{ij}}$,
& $b_{00}=0$\\
\hline
2&$\beta_{22}=(~1,-1,-1,~~1,~~1,~~1)$ &\\
\hline
3& $b_{ij}=\hat{e}.\hat{\beta_{ij}}$,
& $b_{00}=0$,\\
\hline
\end{tabular}
\end{center}
\end{table}
```

To reference the table in the text, code as follows:

```
Table~\ref{tab1} lists the closed-form...
```

to produce:

Table I lists the closed-form . . .

F. Lists

In this section, we will consider three types of lists: simple unnumbered, numbered and bulleted. There have been numerous options added to IEEEtran to enhance the creation of lists. If your lists are more complex than those shown below, please refer to the "IEEEtran_HOWTO.pdf" for additional options.

A plain unnumbered list

bare_jrnl.tex
bare_conf.tex
bare_jrnl_compsoc.tex
bare_conf_compsoc.tex
bare_jrnl_comsoc.tex

coded as:

```
\begin{list}{}{}
\item{bare\_jrnl.tex}
\item{bare\_conf.tex}
\item{bare\_jrnl\_compsoc.tex}
\item{bare\_conf\_compsoc.tex}
\item{bare\_jrnl\_comsoc.tex}
\end{list}
```

A simple numbered list

1) bare_jrnl.tex
2) bare_conf.tex
3) bare_jrnl_compsoc.tex
4) bare_conf_compsoc.tex
5) bare_jrnl_comsoc.tex

coded as:

```
\begin{enumerate}
\item{bare\_jrnl.tex}
\item{bare\_conf.tex}
\item{bare\_jrnl\_compsoc.tex}
\item{bare\_conf\_compsoc.tex}
\item{bare\_jrnl\_comsoc.tex}
\end{enumerate}
```

A simple bulleted list

- bare_jrnl.tex
- bare_conf.tex
- bare_jrnl_compsoc.tex
- bare_conf_compsoc.tex
- bare_jrnl_comsoc.tex

coded as:

```
\begin{itemize}
\item{bare\_jrnl.tex}
\item{bare\_conf.tex}
\item{bare\_jrnl\_compsoc.tex}
\item{bare\_conf\_compsoc.tex}
\item{bare\_jrnl\_comsoc.tex}
\end{itemize}
```

G. Other Elements

For other less common elements such as Algorithms, Theorems and Proofs, and Floating Structures such as page-wide tables, figures or equations, please refer to the "IEEEtran_HOWTO.pdf" section on "Double Column Floats."

IX. How to Create Common Back Matter Elements

The following sections demonstrate common back matter elements such as Acknowledgments, Bibliographies, Appendicies and Author Biographies.

A. Acknowledgments

This should be a simple paragraph before the bibliography to thank those individuals and institutions who have supported your work on this article.

```
\section{Acknowledgments}
\noindent Text describing those who
supported your paper.
```

B. Bibliographies

References Simplified: A simple way of composing references is to use the \bibitem macro to define the beginning of a reference as in the following examples:

[6] H. Sira-Ramirez. "On the sliding mode control of nonlinear systems," Systems & Control Letters, vol. 19, pp. 303–312, 1992.
coded as:

\bibitem{Sira3}
H. Sira-Ramirez. ''On the sliding mode
control of nonlinear systems,''
\textit{Systems \& Control Letters},
vol. 19, pp. 303--312, 1992.

[7] A. Levant."Exact differentiation of signals with unbounded higher derivatives," in Proceedings of the 45th IEEE Conference on Decision and Control, San Diego, California, USA, pp. 5585–5590, 2006.
coded as:

\bibitem{Levant}
A. Levant. ''Exact differentiation of
signals with unbounded higher
derivatives,'' in \textit{Proceedings
of the 45th IEEE Conference on
Decision and Control}, San Diego,
California, USA, pp. 5585--5590, 2006.

[8] M. Fliess, C. Join, and H. Sira-Ramirez. "Non-linear estimation is easy," International Journal of Modelling, Identification and Control, vol. 4, no. 1, pp. 12–27, 2008.
coded as:

\bibitem{Cedric}
M. Fliess, C. Join, and H. Sira-Ramirez.
''Non-linear estimation is easy,''
\textit{International Journal of Modelling,
Identification and Control}, vol. 4,
no. 1, pp. 12--27, 2008.

[9] R. Ortega, A. Astolfi, G. Bastin, and H. Rodriguez. "Stabilization of food-chain systems using a port-controlled Hamiltonian description," in Proceedings of the American Control Conference, Chicago, Illinois, USA, pp. 2245–2249, 2000.
coded as:

\bibitem{Ortega}
R. Ortega, A. Astolfi, G. Bastin, and H.
Rodriguez. ''Stabilization of food-chain
systems using a port-controlled Hamiltonian
description,'' in \textit{Proceedings of the
American Control Conference}, Chicago,
Illinois, USA, pp. 2245--2249, 2000.

C. Accented Characters in References

When using accented characters in references, please use the standard LaTeX coding for accents. Do not use math coding for character accents. For example:

\'e, \"o, \`a, \~e

will produce: é, ö, à, ẽ

D. Use of BibTeX

If you wish to use BibTeX, please see the documentation that accompanies the IEEEtran Bibliography package.

E. Biographies and Author Photos

Authors may have options to include their photo or not. Photos should be a bit-map graphic (.tif or .jpg) and sized to fit in the space allowed. Please see the coding samples below:

\begin{IEEEbiographynophoto}{Jane Doe}
Biography text here without a photo.
\end{IEEEbiographynophoto}

or a biography with a photo

\begin{IEEEbiography}[{\includegraphics
[width=1in,height=1.25in,clip,
keepaspectratio]{fig1.png}}]
{IEEE Publications Technology Team}
In this paragraph you can place
your educational, professional background
and research and other interests.
\end{IEEEbiography}

Please see the end of this document to see the output of these coding examples.

X. Mathematical Typography and Why It Matters

Typographical conventions for mathematical formulas have been developed to provide uniformity and clarity of presentation across mathematical texts. This enables the readers of those texts to both understand the author's ideas and to grasp new concepts quickly. While software such as LATEX and MathType® can produce aesthetically pleasing math when used properly, it is also very easy to misuse the software, potentially resulting in incorrect math display.

IEEE aims to provide authors with the proper guidance on mathematical typesetting style and assist them in writing the best possible article.

As such, IEEE has assembled a set of examples of good and bad mathematical typesetting. You will see how various issues are dealt with. The following publications have been referenced in preparing this material:

Mathematics into Type, published by the American Mathematical Society
The Printing of Mathematics, published by Oxford University Press
The LATEXCompanion, by F. Mittelbach and M. Goossens
More Math into LaTeX, by G. Grätzer
AMS-StyleGuide-online.pdf, published by the American Mathematical Society

Further examples can be seen at http://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE-Math-Typesetting-Guide.pdf

A. Display Equations

A simple display equation example shown below uses the "equation" environment. To number the equations, use the \label macro to create an identifier for the equation. LaTeX will automatically number the equation for you.

$$x = \sum_{i=0}^{n} 2iQ. \tag{3}$$

is coded as follows:

```
\begin{equation}
\label{deqn_ex1}
x = \sum_{i=0}^{n} 2{i} Q.
\end{equation}
```

To reference this equation in the text use the \ref macro. Please see (3)
is coded as follows:

```
Please see (\ref{deqn_ex1})
```

B. Equation Numbering

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end, i.e., (1), (2), (3), (4), (5), etc. Do not use roman numerals or section numbers for equation numbering.

Appendix Equations: The continuation of consecutively numbered equations is best in the Appendix, but numbering as (A1), (A2), etc., is permissible.

Hyphens and Periods: Hyphens and periods should not be used in equation numbers, i.e., use (1a) rather than (1-a) and (2a) rather than (2.a) for sub-equations. This should be consistent throughout the article.

C. Multi-line equations and alignment

Here we show several examples of multi-line equations and proper alignments.
A single equation that must break over multiple lines due to length with no specific alignment.

$$\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline} \tag{4}$$

is coded as:

```
\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline}
```

A single equation with multiple lines aligned at the = signs

$$a = c + d \tag{5}$$
$$b = e + f \tag{6}$$

is coded as:

```
\begin{align}
a &= c+d \\
b &= e+f
\end{align}
```

The align environment can align on multiple points as shown in the following example:

$$x = y \qquad X = Y \qquad a = bc \tag{7}$$
$$x' = y' \qquad X' = Y' \qquad a' = bz \tag{8}$$

is coded as:

```
\begin{align}
x &= y & X & =Y & a &=bc\\
x' &= y' & X' &=Y' &a' &=bz
\end{align}
```

D. Subnumbering

The amsmath package provides a subequations environment to facilitate subnumbering. An example:

$$f = g \tag{9a}$$
$$f' = g' \tag{9b}$$
$$\mathcal{L}f = \mathcal{L}g \tag{9c}$$

is coded as:

```
\begin{subequations}\label{eq:2}
\begin{align}
f&=g \label{eq:2A}\\
f' &=g' \label{eq:2B}\\
\mathcal{L}f &= \mathcal{L}g \label{eq:2c}
\end{align}
\end{subequations}
```

E. Matrices

There are several useful matrix environments that can save you some keystrokes. See the example coding below and the output.
A simple matrix:

$$\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \tag{10}$$

is coded as:

```
\begin{equation}
\begin{matrix}  0 &  1 \\
1 &  0 \end{matrix}
\end{equation}
```

A matrix with parenthesis

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{11}$$

is coded as:

```
\begin{equation}
```

```
\begin{pmatrix} 0 & -i \\
 i &  0 \end{pmatrix}
\end{equation}
```

A matrix with square brackets

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{12}$$

is coded as:

```
\begin{equation}
\begin{bmatrix} 0 & -1 \\
1 &  0 \end{bmatrix}
\end{equation}
```

A matrix with curly braces

$$\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix} \tag{13}$$

is coded as:

```
\begin{equation}
\begin{Bmatrix} 1 &  0 \\
0 & -1 \end{Bmatrix}
\end{equation}
```

A matrix with single verticals

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \tag{14}$$

is coded as:

```
\begin{equation}
\begin{vmatrix} a &  b \\
c &  d \end{vmatrix}
\end{equation}
```

A matrix with double verticals

$$\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix} \tag{15}$$

is coded as:

```
\begin{equation}
\begin{Vmatrix} i &  0 \\
0 & -i \end{Vmatrix}
\end{equation}
```

F. Arrays

The array environment allows you some options for matrix-like equations. You will have to manually key the fences, but you'll have options for alignment of the columns and for setting horizontal and vertical rules. The argument to array controls alignment and placement of vertical rules.

A simple array

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \tag{16}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

A slight variation on this to better align the numbers in the last column

$$\left(\begin{array}{cccr} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \tag{17}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccr}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

An array with vertical and horizontal rules

$$\left(\begin{array}{c|c|c|r} a+b+c & uv & x-y & 27 \\ \hline a+b & u+v & z & 134 \end{array} \right) \tag{18}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{c|c|c|r}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

Note the argument now has the pipe "|" included to indicate the placement of the vertical rules.

G. Cases Structures

Many times we find cases coded using the wrong environment, i.e., array. Using the cases environment will save keystrokes (from not having to type the \left\lbrace) and automatically provide the correct column alignment.

$$z_m(t) = \begin{cases} 1, & \text{if } \beta_m(t) \\ 0, & \text{otherwise.} \end{cases}$$

is coded as follows:

```
\begin{equation*}
{z_m(t)} =
\begin{cases}
1,&{\text{if}}\ {\beta }_m(t),\\
{0,}&{\text{otherwise.}}
\end{cases}
\end{equation*}
```

Note that the "&" is used to mark the tabular alignment. This is important to get proper column alignment. Do not use \quad or other fixed spaces to try and align the columns. Also, note the use of the \text macro for text elements such as "if" and "otherwise".

H. Function Formatting in Equations

In many cases there is an easy way to properly format most common functions. Use of the \ in front of the function name will in most cases, provide the correct formatting. When this does not work, the following example provides a solution using the \text macro.

$$d_R^{KM} = \underset{d_l^{KM}}{\text{arg min}} \{ d_1^{KM}, \ldots, d_6^{KM}\}.$$

is coded as follows:

```
\begin{equation*}
 d_{R}^{KM} = \underset {d_{l}^{KM}}
 {\text{arg min}} \{ d_{1}^{KM},
 \ldots,d_{6}^{KM}\}.
\end{equation*}
```

I. Text Acronyms inside equations

This example shows where the acronym "MSE" is coded using \text{} to match how it appears in the text.

$$\text{MSE} = \frac {1}{n}\sum _{i=1}^{n}(Y_{i} - \hat {Y_{i}})^{2}$$

```
\begin{equation*}
 \text{MSE} = \frac {1}{n}\sum _{i=1}^{n}
(Y_{i} - \hat {Y_{i}})^{2}
\end{equation*}
```

J. Obsolete Coding

Avoid the use of outdated environments, such as eqnarray and $$ math delimiters, for display equations. The $$ display math delimiters are left over from PlainTeX and should not be used in LaTeX, ever. Poor vertical spacing will result.

K. Use Appropriate Delimiters for Display Equations

Some improper mathematical coding advice has been given in various YouTube™ videos on how to write scholarly articles, so please follow these good examples:

For single-line unnumbered display equations, please use the following delimiters:

\[. . . \] or

\begin{equation*} . . . \end{equation*}

Note that the * in the environment name turns off equation numbering.

For multiline unnumbered display equations that have alignment requirements, please use the following delimiters:

\begin{align*} . . . \end{align*}

For single-line numbered display equations, please use the following delimiters:

\begin{equation} . . . \end{equation}

For multiline numbered display equations, please use the following delimiters:

\begin{align} . . . \end{align}

XI. LaTeX Package Suggestions

Immediately after your documenttype declaration at the top of your LaTeX file is the place where you should declare any packages that are being used. The following packages were used in the production of this document.

```
\usepackage{amsmath,amsfonts}
\usepackage{algorithmic}
\usepackage{array}
\usepackage[caption=false,font=normalsize,
   labelfont=sf,textfont=sf]{subfig}
\u00sepackage{textcomp}
\usepackage{stfloats}
\usepackage{url}
\usepackage{verbatim}
\usepackage{graphicx}
\usepackage{balance}
```

XII. Additional Advice

Please use "soft" (e.g., \eqref{Eq}) or (\ref{Eq}) cross references instead of "hard" references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please note that the {subequations} environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you've discovered a new method of counting.

BibTeX does not work by magic. It doesn't get the bibliographic data from thin air but from .bib files. If you use BibTeX to produce a bibliography you must send the .bib files.

LaTeX can't read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

LaTeX does not have precognitive abilities. If you put a \label command before the command that updates the counter it's supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a \label command should not go before the caption of a figure or a table.

Please do not use \nonumber or \notag inside the {array} environment. It will not stop equation numbers inside {array} (there won't be any anyway) and it might stop a wanted equation number in the surrounding equation.

XIII. A Final Checklist

1) Make sure that your equations are numbered sequentially and there are no equation numbers missing or duplicated. Avoid hyphens and periods in your equation numbering. Stay with IEEE style, i.e., (1), (2), (3) or for sub-equations (1a), (1b). For equations in the appendix (A1), (A2), etc..
2) Are your equations properly formatted? Text, functions, alignment points in cases and arrays, etc.
3) Make sure all graphics are included.
4) Make sure your references are included either in your main LaTeX file or a separate .bib file if calling the external file.

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Jane Doe Biography text here without a photo.

IEEE Publications Technology Team In this paragraph you can place your educational, professional background and research and other interests.